# GATE-AII BRANCHES
# Engineering Mathematics




## Probability and Statistics

Lecture No.- 12



By- Rahul sir

# Recap of Previous Lecture



**Topic** Expectation,variance and standard deviation .

# Topics to be Covered



**Topic** Concept of probability distribution

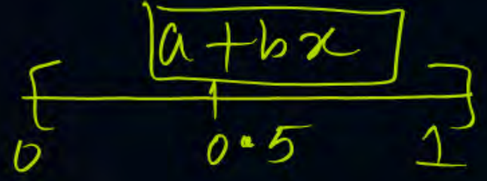
**Topic** Concept of Binomial distribution

**Topic** Problems based on Binomial distribution

- Understand The Problem
- Unknown
- Known
- Link
- Devise The Plan
- Soln



# Topic :Probability and statistics




#Q. A random variable X has probability density function f(x) as given below:

$$f(x)=\begin{cases} a+bx & \text{for } 0<x<1 \\ 0 & \text{otherwise} \end{cases}$$

$\int_0^1 (a+bx)\,dx = 1$ → a, b

If the expected value E[X] = 2/3 then Pr[X < 0.5] is_____.

$E[X]=\frac{2}{3}$ → a, b.

$P[X<0.5]=\int_0^{0.5}(a+bx)\,dx$ → [a unknown, b unknown]

$$E[X] = \frac{2}{3}$$

$$\text{Expectation} = \int_a^b x f(x)\,dx = \frac{2}{3}$$

$$\Rightarrow \int_0^1 x(a+bx)\,dx = \frac{2}{3}$$

$$\Rightarrow \left[\frac{ax^2}{2} + \frac{bx^3}{3}\right]_0^1 = \frac{2}{3}$$

$$\Rightarrow \boxed{\frac{a}{2} + \frac{b}{3} = \frac{2}{3}} \quad -(1)$$

$$\frac{a}{2} + \frac{b}{2} = 1 \quad -(1)$$

$$\frac{a}{2} + \frac{b}{3} = \frac{2}{3} \quad -(2)$$

$$a = 0, \quad b = 2$$

Valid pdf property

$$\int_a^b f(x)\,dx = 1$$

$$\int_0^1 (a+bx)\,dx = 1$$

$$\left[ax + \frac{bx^2}{2}\right]_0^1 = 1$$

$$\Rightarrow \boxed{a + \frac{b}{2} = 1} \quad -(2)$$

$$P(X<0.5) = \int_0^{0.5} (a+bx)\,dx$$

$$= \int_0^{0.5} (0+2x)\,dx$$

$$= \left[\frac{x^2}{2}\cdot 2\right]_0^{.5}$$

$$= 0.25$$

$$= \frac{1}{4}$$ ✓

$$\Rightarrow \lim_{h\to 0} \int_0^{0.5-h} (a+bx)\,dx$$

$$\Rightarrow \lim_{h\to 0} \int_{0+h}^{0.5} (a+bx)\,dx$$



# Topic :Probability and statistics

#Q. A fair die with faces {1, 2, 3, 4, 5, 6} is thrown repeatedly till '3' is observed for the first time. Let X denote the number of times the dice is thrown. The expected value of X is_____.

X = No of Times die thrown

expected value of X

$X$ = No of Die is Thrown

(Discrete Random variable)

$X = 1$ success must — $S$ $\quad = \frac{1}{6}$

$X = 2$ F S stop $\quad = \overline{3}\,3 \quad = \frac{5}{6} \times \frac{1}{6}$

$X = 3$ FF S stop $\quad = \overline{3}\,\overline{3}\,3 \quad = \left(\frac{5}{6}\right)^2 \frac{1}{6}$

$X = 4$ FFF S $\quad = \overline{3}\,\overline{3}\,\overline{3}\,3 \quad = \left(\frac{5}{6}\right)^3 \frac{1}{6}$

$X = 5$ FFFF S $\quad = \overline{3}\,\overline{3}\,\overline{3}\,\overline{3}\,3 \quad = \left(\frac{5}{6}\right)^4 \cdot \frac{1}{6}$

$X = 6$ FFFFF S $\quad = \overline{3}\,\overline{3}\,\overline{3}\,\overline{3}\,\overline{3}\,3 \quad = \left(\frac{5}{6}\right)^5 \cdot \frac{1}{6}$

Infinite countable SET

$P(\text{till } 3) = \text{success}$

$P(3) = \frac{1}{6}$

$P(\overline{3}) = \frac{5}{6}$

Geometric random variable

$\frac{5}{6}, \left(\frac{5}{6}\right)^2, \left(\frac{5}{6}\right)^3$ $T_1\ T_2\ T_3$ $r = \frac{T_2}{T_1} = \frac{5}{6}$

| $X$ | 1 | 2 | 3 | 4 | 5 | 6 | — — — |
|---|---|---|---|---|---|---|---|
| $P(X=x)$ | $\frac{1}{6}$ | $\frac{5}{6}\cdot\frac{1}{6}$ | $\left(\frac{5}{6}\right)^2\frac{1}{6}$ | $\left(\frac{5}{6}\right)^3\frac{1}{6}$ | $\left(\frac{5}{6}\right)^4\cdot\frac{1}{6}$ | $\left(\frac{5}{6}\right)^5\frac{1}{6}$ | — — — |

Expectation of random variable $E[X] = \sum_{i=1}^{n} x_i p_i$

A · P
G · P ] AGP

$$E[X] = 1\cdot\frac{1}{6} + 2\left(\frac{5}{6}\right)\left(\frac{1}{6}\right) + 3\left(\frac{5}{6}\right)^2\frac{1}{6} + 4\left(\frac{5}{6}\right)^3\frac{1}{6} + 5\left(\frac{5}{6}\right)^4\frac{1}{6} + \text{— —}$$

G.P ] Common ratio SERIES multiply

shifting →

$$\frac{5}{6}E[X] = \quad + \left(\frac{5}{6}\right)\frac{1}{6} + 2\left(\frac{5}{6}\right)^2\frac{1}{6} + 3\left(\frac{5}{6}\right)^3\frac{1}{6} + 4\left(\frac{5}{6}\right)^4\frac{1}{6} + \text{— —}$$

$$\frac{1}{6}E[X] = 1\cdot\frac{1}{6} + \left(\frac{5}{6}\right)\left(\frac{1}{6}\right) + \left(\frac{5}{6}\right)^2\frac{1}{6} + \left(\frac{5}{6}\right)^3\frac{1}{6} + \frac{1}{6}\left(\frac{5}{6}\right)^4 + \text{— —}$$

$$\frac{1}{6}E[X] = \frac{1}{6} + \frac{1}{6}\left[\frac{5}{6} + \left(\frac{5}{6}\right)^2 + \left(\frac{5}{6}\right)^3 + \text{—}\right]$$

SUM $= \frac{a}{(1-r)}$ $r = \frac{T_2}{T_1}$ $r = 5/6$

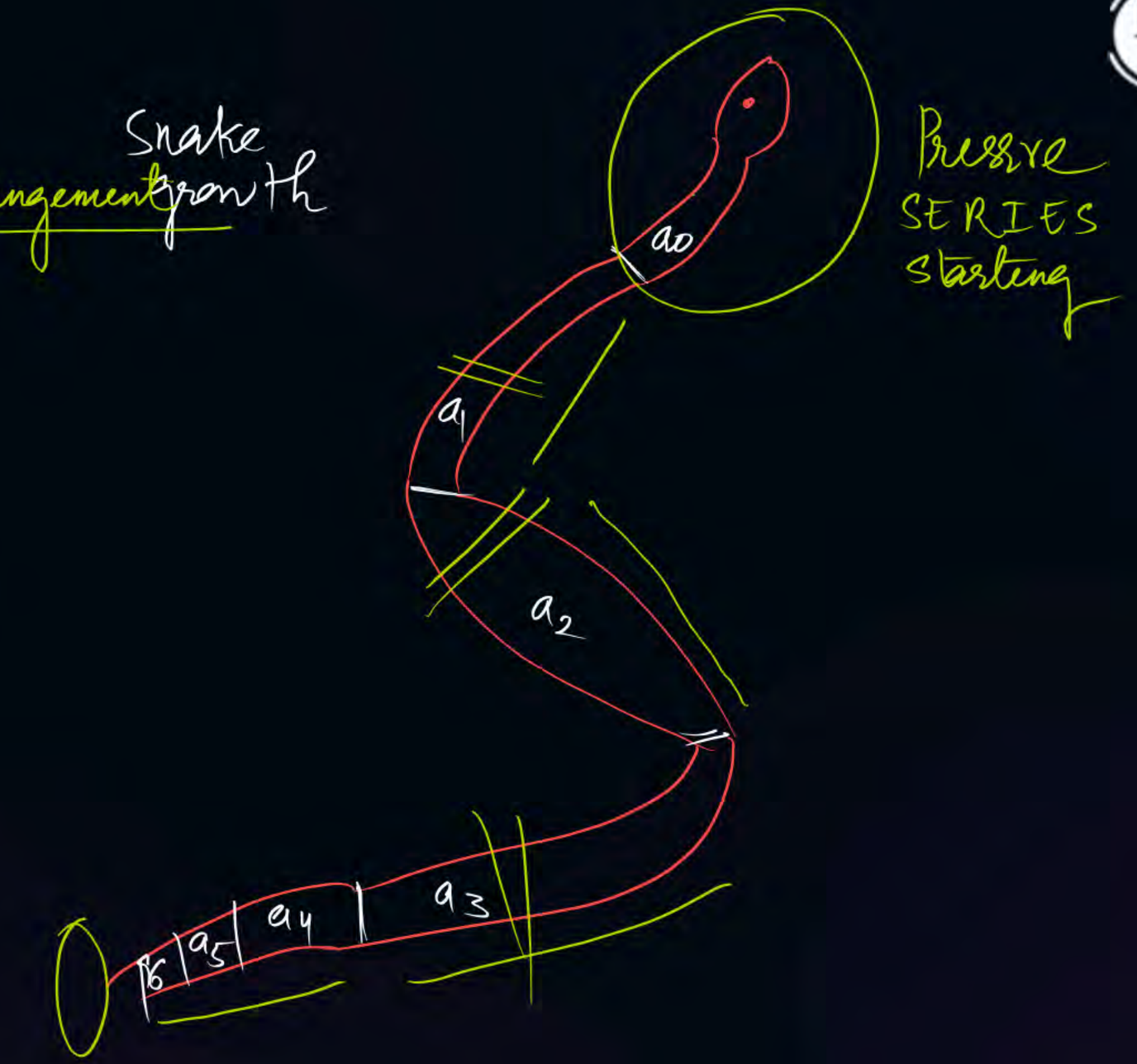
$$\frac{1}{6}E[X] = \frac{1}{6} + \left[\frac{\frac{5}{6}}{1-5/6}\right]$$

$$\boxed{E[X] = 6}$$ Ans

$v_1 - v_2$
$v_2 - v_3$
$v_3 - v_4$
$v_5 - v_{14}$

reassignment growth

Snake




Pressure SERIES starting



# Topic :Probability and statistics

$V(X) = E[X^2] - [E[X]]^2$

#Q. The variance of the random variable X with probability density function $f(x) = \frac{1}{2}|x|e^{-|x|}$ is_____.

$f(x) = \frac{1}{2}|x|e^{-|x|}$




$$E[x^2] \Rightarrow \int_{-\infty}^{\infty} x^2 f(x)\,dx \Rightarrow \int_{-\infty}^{\infty} x^2 \cdot \frac{1}{2}|x|e^{-|x|}\,dx$$

$$\Rightarrow \int_{-\infty}^{0} x^2 \frac{1}{2}(-x)e^{x}\,dx + \int_{0}^{\infty} x^2 \cdot \frac{1}{2}(x)e^{-x}\,dx$$

$$E[x^2] \Rightarrow \int_{-\infty}^{0} \frac{-x^3}{2}e^{x}\,dx + \int_{0}^{\infty} \frac{x^3}{2}e^{-x}\,dx = 6$$

$$E[X] = \int_{-\infty}^{\infty} x f(x)\,dx$$

$$= \int_{-\infty}^{0} x \cdot \frac{1}{2}(-x) e^{x}\,dx + \int_{0}^{\infty} x \cdot \frac{1}{2}(x) e^{-x}\,dx$$

$$E[X] = \int_{-\infty}^{0} \left(-\frac{x^2}{2}\right) e^{x}\,dx + \int_{0}^{\infty} \frac{x^2}{2} e^{-x}\,dx = 0$$

✓ Short cut (Integration via Parts)

$$I = \int \underbrace{x^2}_{D} \underbrace{e^{x}}\,dx$$

→ algebraic function — derivative

→ exponential function — integration

Algebraic Function must

$$V(X) = E[X^2] - [E[X]]^2 = 6 - 0 = 6$$

$$I = \int x^2 e^x$$

| D | | I |
|---|---|---|
| $x^2$ | + | $e^x$ |
| $2x$ | − | $e^x$ |
| $2$ | + | $e^x$ |
| $0$ | | $e^x$ |

becomes ZERO stop.

$$= x^2e^x - 2xe^x + 2e^x$$

$$I = \int x^2 \cos x \, dx$$

| | I |
|---|---|
| $x^2$ | + $\cos x$ |
| $2x$ | − $+\sin x$ |
| $2$ | + $-\cos x$ |
| $0$ | $-\sin x$ |

$$I = x^2\sin x + 2x\cos x - 2\sin x$$

$$I = \int x^3 e^{-2x} \, dx$$

| | | |
|---|---|---|
| $x^3$ | + | $e^{-2x}$ |
| $3x^2$ | − | $(e^{-2x})/-2$ |
| $6x$ | + | $e^{-2x}/4$ |
| $6$ | − | $e^{-2x}/-8$ |
| $0$ | | $e^{-2x}/16$ |

Ans

# Imp Probability Distribution

Discrete Distributions + continuous dist

## Discrete Distribution : Arrival Pattern

### Bernoulli Distribution : Bernoulli Trials

- Tossing A single coin → H (Head) S, T (Tail) F → (Arrival)
- Throwing A Die → coming 6 S, Not coming 6 F
-  → white ball (S), Red ball (F)

Bernoulli Trials (Independent)

Single Trial

Bernoulli Trials

- Success (HEAD) (1)
- Failure (Tail) (0)

Indicator Function

$X = 0, 1$

Tossing A coin
- H present (1)
- T (Not present 0)

$X$ = Bernoulli Random variable

only Two values

$X = 0, 1$

Prob of success $= p$

Prob of Failure $= q = (1-p)$

| $X$ | 0 | 1 |
|---|---|---|
| $P(X=x)$ | $(1-p)$ | $(p)$ |

$$P[X=x] = p^x (1-p)^{1-x}$$

$$P[X=x] = p^x (1-p)^{1-x}$$

$$\begin{cases} P[X=0] = p^0 (1-p)^{1-0} = (1-p) \\ P[X=1] = p^1 (1-p)^{1-1} = p \end{cases}$$

## Statistical Parameters:

| X | 0 | 1 |
|---|---|---|
| P(X=x) | 1-p | p |

# MEAN $= E[X] = \sum_{i=1}^{n} x_i p_i$

$$E[X] = 0 \times (1-p) + 1 \times p = p$$

$$\boxed{E[X] = p}$$

# Variance $= E[X^2] - [E[X]]^2$

$$\Rightarrow (0)^2 \times (1-p) + (1)^2 \times p - (p^2)$$

$$= p - p^2 = p(1-p)$$

$$\boxed{\text{Var}(X) = p(1-p)}$$

Standard deviation $= \sqrt{\text{variance}}$

$= \sqrt{p(1-p)}$

| 0 | 1 |
|---|---|

(Bernoulli Trials always independent)




] No. of Trials Are Fixed

No. of Trials Are fixed

Prob (S) + prob (F) = 1

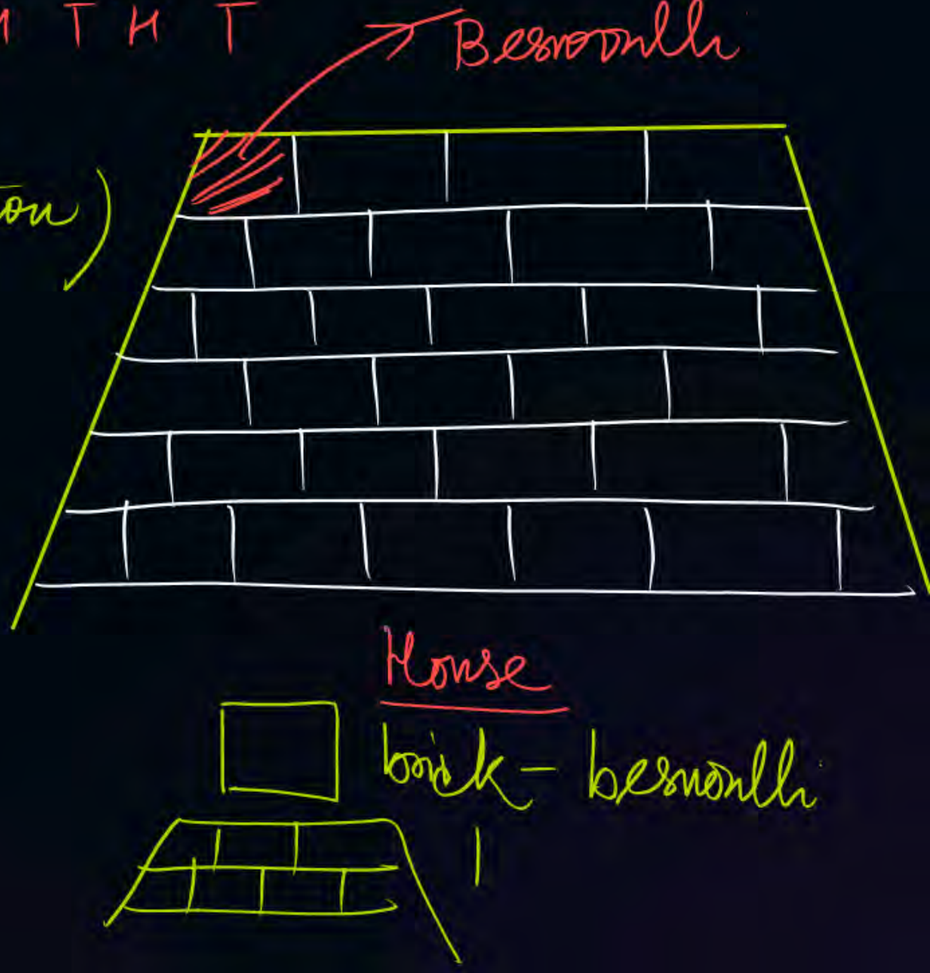
(Binomial Distribution)

Prob. of Success = p

Prob. of failure = q

No. of trials = n

No. of Success = (r)



prob of success = $r$ times

prob of Failure = $(n-r)$ times

prob of success = $p \cdot p \cdot p \cdot p \cdot p$ --- $r$ times (Independent)

prob of Failure = $q\ q\ q\ q\ q$ --- $(n-r)$ times

$$P[X = r \text{ success}] = {}^{n}C_{r}\, p^{r}\, q^{n-r}$$

success = Arrival

Bionomial Distribution

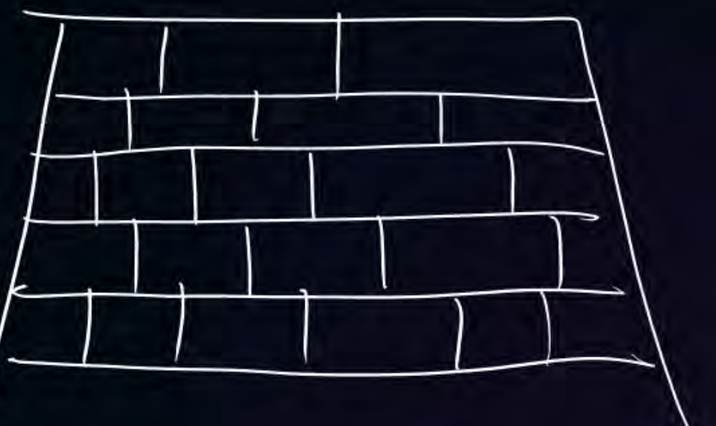


Bernoull Distribution

$B(p)$

Binomial Distribution

$B(n, p, r)$



$$B(n,p) = P(X=r) = {}^{n}C_{r}\, p^{r} q^{n-r}$$

✓ Every time Trial are fixed
✓ prob. fixed
✓ $r$ = vary

⇒ Statistical Averages

| | Bernoulli (single) | Binomial ($n$) |
|---|---|---|
| Arrival Pattern | MEAN = $p$ | mean = $np$ |
| | variance = $p(1-p) = pq$ | variance = $np(1-p) = npq$ |
| | Standard deviation = $\sqrt{pq}$ | S.D = $\sqrt{npq}$ |
| | Parametres = (1) | binomial → $(n, p)$ |
| |   |  |

$1-p = q$

Arrival Pattern

# THANK - YOU